

JeongHyeon Kim

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RESEARCH INTEREST

My research interests lie in **humanoid robotics** and **whole-body control**. I am interested in developing controllers that enable humanoid robots to perform complex, dynamic tasks through the integration of model-based and learning-based approaches.

EDUCATION

- Korea University**
M.S. in Mechanical Engineering
Mar.2025 – Present
Seoul, Republic of Korea
- Korea University**
B.S. in Mechanical Engineering
2019 – 2025
Seoul, Republic of Korea

EXPERIENCE

- Human-Machine Systems Lab, Korea University**
Graduate Research Student (M.S.)
Mar.2025 – Present
Seoul, Republic of Korea
– Research on humanoid robotics and whole-body control
- Human-Machine Systems Lab, Korea University**
Undergraduate Research Student
Mar.2024 – Feb.2025
Seoul, Republic of Korea
– Participated in robotics research projects

PROJECTS

- Quadruped Robot Obstacle Avoidance (CCP) — Team Leader**
Computer Vision, Autonomous Navigation
Mar.2024 – Jan.2025
– Developed obstacle detection and avoidance pipeline for DOGZILLA quadruped robot
– Implemented monocular ORB-SLAM for depth estimation and UV-disparity for obstacle segmentation
– Awarded President’s Award, Creative Challenger Program (CCP), Korea University
- Ladder-Crossing Robot — Team Leader**
Mechanism Design and Control
Mar.2023 – Sep.2023
– Tools: Arduino, Creo, NX
– Programming Language: C++
- Solar Tracking Parasol — Team Leader**
Mechanism Design and Control
Sep.2023 – Nov.2023
– Tools: Arduino, AutoCAD
– Programming Language: C++
– Bronze Award, Creative Comprehensive Design Competition, Korea University

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB
Robotics: ROS2, Gazebo, URDF/XACRO
Tools: OpenCV, Arduino, Creo, NX, AutoCAD, SolidWorks, Git
OS: Linux (Ubuntu)

ACHIEVEMENTS

- President’s Award**Creative Challenger Program (CCP), Korea University
Jan.2025
- Selected**Master’s Research Encouragement Grant, National Research Foundation of Korea
2025
- Bronze Award**Creative Comprehensive Design Competition, Korea University
Sep.2023
- Awarded**Capstone Design Fair, Korea University
Nov.2023